

Markscheme

Mathematics: analysis and approaches

Practice question bank

11. (a)

Given that $3^x = 5$.

Find the value of 3^{2x+1} .

$$3^{2x+1} = 3^{2x} \times 3 = (3^x)^2 \times 3 \quad (M1)$$

$$\text{substitute } 3^x = 5 \quad (M1)$$

$$= 5^2 \times 3 = 25 \times 3 \quad \mathbf{A1}$$

$$= 75 \quad \mathbf{A1}$$

[4 marks]

(b)

Hence find the value of 3^{4x+2} .

$$3^{4x+2} = (3^{2x+1})^2 \quad (M1)$$

$$= 75^2 = 5625 \quad \mathbf{A1}$$

[2 marks]